Gravity on the 7+1 spacetime: provenance

Sources, decisions, derivations and the verification record of the curvature, the Einstein and Lovelock tensors, the generalized frame matrices, the canonical spin connection, the covariant Dirac equation of dirac16complex and the three cosmological epochs — sections 11–14 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

This record states what was asked, what was used, every decision taken with the equation behind it, how each result was derived and how it was checked: the 97 checks of sections 11–14 of 7+1.nb (97 True), the 99 boolean checks of the independent Python program tools/ref_gravity.py (99 True), the comparison of the printed numbers, the program versions and the SHA-256 hashes of the files.

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1. The task

This stage of the project asked for the following (summarized; the original wording is the author's private working material and is not reproduced):

1. Take the IDEAS of the repository github.com/once-ere/Dirac_claude, which is cast on a 4+4 spacetime with four time-like coordinates and a definite warped metric, and rebuild them on the 7+1 spacetime of this project: ONE time-like coordinate x_0 and SEVEN space-like coordinates $x_1, \ldots, x_7$. The 4+4 metric of Dirac_claude is not used.
2. Einstein and Lovelock tensors of 7+1 spacetime.
3. Generalized frame matrices, a non-trivial canonical spin connection that couples gravity to dirac16complex, and a total covariant derivative under which the frame matrices are constant.
4. Kohn–Sham theory of dirac16complex in gravitational fields (primordial near $x_0 = 0$, modern, far future), its self-coupling, and the totality of the Standard Model (sections 15–19 of 7+1.nb; this record covers sections 11–14).
5. Compute the Lovelock tensors as in the author's reference notebook Einstein-Lovelock-7+1-Nash.nb and do NOT use its section "FAILED Model metric g71; DO NOT USE!".

6. For every documentation file a companion provenance file and a symbol file, each in Markdown and LaTeX, the LaTeX compiled to PDF; no document refers the reader to another one, and every equation used is written in the document that uses it.

7. Push every completed stage to github.com/once-ere/7plus1, respecting the repository's .gitignore; use only the Wolfram front end WolframNB.exe of Wolfram 15.0.1.

The coordinates of 7+1.nb are kept: $X = (x_0, \ldots, x_7)$, x_0 the time (the direction of the octonion unit), x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space; metric signature $(+, -, \ldots, -)$ with $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$.

2. Inputs

2.1 Objects of the earlier sections of 7+1.nb

Sections 11–14 use, unchanged, the coordinates `X = {x0, …, x7}`, the metric `η17` $= \eta$, the identity and zero matrices `ID16`, `Zero16`, the check registry `check[name, test]`, and the real 16×16 Dirac matrices `Γ17[A]` of section 4 with

$$\Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix} \ (i = 1, \ldots, 7), \qquad \Gamma_A \Gamma_B + \Gamma_B \Gamma_A = 2\eta_{AB} 1_{16}, \qquad \Gamma^A = \eta^{AA} \Gamma_A,$$

where the real 8×8 matrices τ_i^O represent the seven imaginary octonion units ($(\tau_i^O)^2 = -1_8$, $\tau_i^O \tau_j^O + \tau_j^O \tau_i^O = -2\delta_{ij} 1_8$), and the Hamiltonian matrices `αD[j]` $= \Gamma^0 \Gamma^j$, `βD` $= \Gamma_0$ of section 7. Before sections 11 and following were appended, the evaluated notebook was copied to `backups/7+1_2026-10-26_before_gravity.nb` (excluded from the repository by .gitignore).

```
backups/7+1_2026-10-26_before_gravity.nb
604743 bytes
SHA-256 438d972073e2fef9d7da6d57a541432b0fb1ea08424307f59ae2f3f7752bbfaf
```

2.2 The reference notebook Einstein-Lovelock-7+1-Nash.nb

The author's notebook Einstein-Lovelock-7+1-Nash.nb (P. L. Nash, 2022, GNU GPL; corrected version of 27 October 2026, with its PDF printout) shows how the Lovelock tensors are to be computed. It is the author's file and is not part of the repository. Taken from it, unchanged in substance:

- the xAct packages xCoba and xTras, the manifold `M8` with the abstract indices a, b, c, d, f, i, j, k, l, m, n, p, q, r, s, y, the chart `chartM8` with coordinate scalars `x0[]`, …, `x7[]`, and `MyArrayComponents`;
- the generalized Kronecker delta `kδ[lower_, upper_] := Det[Outer[delta, lower, upper]]`;
- the three Lovelock tensors with their normalization,

$$\mathrm{Lovelock}_k = -\frac{1}{2^{k+1}} R_{c_1 d_1}{}^{f_1 i_1} \cdots R_{c_k d_k}{}^{f_k i_k} \, \delta_{a\, f_1 i_1 \cdots f_k i_k}^{b\, c_1 d_1 \cdots c_k d_k} \quad (k = 1, 2, 3), \qquad \mathrm{Lovelock}_1 = R_a{}^b - \tfrac{1}{2}\delta_a{}^b R,$$

- the evaluation of components by assigning a component metric and setting `CD = LC[g]`, `RiemannCD = Riemann[CD]`, `RicciCD = Ricci[CD]`, `RicciScalarCD = RicciScalar[CD]`, `EinsteinCD = Einstein[CD]`;
- the form of the vacuum field equations and the procedure of solving them for w_2, w_3 (and Λ):

$$0 = -\Lambda\, \delta_\nu^\mu + H^{-2} w_1 \,\mathrm{Lovelock}_1 + H^{-4} w_2 \,\mathrm{Lovelock}_2 + H^{-6} w_3 \,\mathrm{Lovelock}_3,$$

with the justification it quotes from D. Lovelock and H. Rund, *Tensors, Differential Forms, and Variational Principles* (Dover), Remark 2 after Eq. (4.38): in n dimensions the most general symmetric divergence-free tensor built from the metric and its first two derivatives is

$$A^{lh} = \sqrt{|g|}\sum_{k=1}^{m-1} \alpha_{(k)}\, g^{jl}\, \delta^{hh_1\cdots h_{2k}}_{jj_1\cdots j_{2k}}\, R^{j_1j_2}{}_{h_1h_2}\cdots R^{j_{2k-1}j_{2k}}{}_{h_{2k-1}h_{2k}} + \lambda\sqrt{|g|}\, g^{lh}, \qquad m-1 = \frac{8}{2} - 1 = 3 \;\; (n=8).$$

NOT taken: the section "FAILED Model metric g71; DO NOT USE!" of the reference notebook (its model metric and the solution found for it); its author states that it never worked for the 7+1 signature. No metric of that section appears in this project. Changed: the signature declaration `DefMetric[{7, 1, 0}, …]` (the mostly-plus convention) became `DefMetric[{1, 7, 0}, …]` to match $\eta = \mathrm{diag}(+1,-1,\ldots,-1)$ of this notebook (the declaration does not enter the Lovelock tensors with mixed indices, and both declarations give the same sign of the determinant, $(-1)^q = -1$ for $q = 7$ and for $q = 1$, as a separate kernel with xTensor 1.2.0 confirmed: `SignDetOfMetric` is -1 for a metric declared `{1, 7, 0}` and for one declared `{7, 1, 0}`); the metric symbol is called `g71` as in the reference notebook.

```
Einstein-Lovelock-7+1-Nash.nb   SHA-256 b2dcb7b9f435d644da085fdd9e738a0cc80d3268ef28606ff9f367ed
    ...6828fdfc
Einstein-Lovelock-7+1-Nash.pdf  SHA-256 6580f9fcaff9c7f1337df35a7fce6b4b04c022d0e2d68954e8b3954b
    ...30f45125
```

2.3 Software and other sources

- Wolfram Language 15.0.1 (the front end WolframNB.exe in `C:\Program Files\Wolfram Research\Wolfram\15.0.1`).
- xAct (J. M. Martín-García, D. Yllanes, T. Nutma and others; GPL; www.xact.es), versions as printed by section 12.2 of 7+1.nb: xPerm 1.2.3, xTensor 1.2.0, xCoba 0.8.6, xTras 1.4.2.
- Python 3.13.17 with sympy 1.14.0, numpy 2.5.3, scipy 1.18.1, mpmath 1.3.0 (the independent programs).
- Ideas (no formulas or metrics) from the repository github.com/once-ere/Dirac_claude: a 16-component field coupled to gravity by the canonical spin connection, Einstein–Lovelock gravity, a primordial pair-creation epoch, a Kohn–Sham treatment, the Standard Model on a lower-dimensional world surface.
- The author's private notes on vielbeins in n dimensions (Cartan's structure equation, a static metric with a radial coordinate, the Dirac equation, the Noether current and the U(1) coupling): their 7+1 versions were derived and checked anew (section 4 below gives the derivations, section 5 the checks). The notes are private and not part of the repository.
- Standard textbook results, each derived and checked here: the Landau–Lifshitz curvature conventions, the Lanczos tensor, Kasner solutions, the Gibbons–Hawking temperature $H/(2\pi)$, the Landau–Zener formula.

3. Decisions

#	decision	reason / equation
1	Christoffel symbols are written $\{^{\lambda}_{\mu\nu}\}$	Γ is reserved for the Dirac matrices
2	Riemann tensor of Landau and Lifshitz: $R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu\{^{\rho}_{\sigma\nu}\} - \partial_\nu\{^{\rho}_{\sigma\mu}\} + \{^{\rho}_{\mu\lambda}\}\{^{\lambda}_{\sigma\nu}\} - \{^{\rho}_{\nu\lambda}\}\{^{\lambda}_{\sigma\mu}\}$, $R_{\sigma\nu} = R^{\rho}{}_{\sigma\rho\nu}$	with $(+,-,\ldots,-)$: $G^{\mu}{}_{\nu} - \bar\Lambda\delta^{\mu}_{\nu} = \kappa T^{\mu}{}_{\nu}$ with $T^0{}_0 > 0$; agrees with xAct (the reference notebook's Lovelock$_1$ equals this G)
3	Lovelock tensors normalized as in the reference notebook, $E^{(k)} = -2^{-(k+1)}\delta RR\cdots$	$E^{(1)} = G$, $E^{(2)}$ = Lanczos tensor
4	two forms of the field equations: w-form (reference notebook) and α-form	$\alpha_2 = w_2/(w_1H^2)$, $\alpha_3 = w_3/(w_1H^4)$, $\bar\Lambda = \Lambda H^2/w_1$
5	xAct signature declaration $\{1,7,0\}$	one plus sign (time x_0), seven minus signs, as η

#	decision	reason / equation
6	xAct runs in the notebook kernel; the cell is guarded (`If[! MemberQ[$Manifolds, M8], …]`, `Unset` of the component assignments); `tools/run_eval.wl` evaluates the notebook in a fresh kernel	the cell can be evaluated twice while the notebook stays open; xAct keeps its definitions in the kernel, while a reopened notebook gets a new private context
7	`xActLovelock` returns the Transpose of `MyArrayComponents`	the array has the lower index a first; the pp-wave ($E^0{}_1 = -E^1{}_0$) fixes the order
8	a second, independent computation (pair-set sums) is kept	exact agreement with xAct is a strong check; it also works on exact rational tables
9	the inhomogeneous metric is an explicit polynomial, checked exactly at $x_0 = 1/2$, $x_4 = 1/3$	symbolic checks with arbitrary functions of two variables took hours
10	exponential expansion is evaluated from the xAct components of the cosmological metric (`xlCos`), not by a new xAct run	xAct's final Simplify with symbolic exponentials is very slow
11	the spinor connection is built as `Zero16 + Sum[…]`	a sum of zeros would otherwise be the scalar 0, not a 16×16 matrix
12	the 16-component mode equation is reduced with $\mathcal{Z} = \mathcal{A}\mathcal{B}\beta$, $\mathcal{Z}^2 = -1$	two classes of states ($\pm m\sigma_3$), created at different rates
13	pair creation is integrated from $10^{-15}\lambda$ to 300λ (Kasner), from $-\ln 2000$ to $\ln(10^6/M^2)$ (de Sitter)	adiabatic before and after; checked by $\lvert\alpha\rvert^2 + \lvert\beta\rvert^2 = 1$ and the Landau–Zener formula

4. Derivations

Sectional curvatures of the cosmological metric. In the frame $e = \mathrm{diag}(1, a, a, a, b, b, b, b)$ every $R^{AB}{}_{CD}$ is diagonal on index pairs, with

$$K_{0i} = -(H_a' + H_a^2), \quad K_{0k} = -(H_b' + H_b^2), \quad K_{ij} = -H_a^2, \quad K_{kl} = -H_b^2, \quad K_{ik} = -H_a H_b,$$

In $L_k = 2^{-k}\delta^{\alpha_1\beta_1\cdots}_{\gamma_1\delta_1\cdots} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots$ every ordered sequence of $2k$ different indices contributes $2^k\prod_i K_{\alpha_i\beta_i}$ (each lower pair is the upper pair in one of its two orders), so L_k is a sum over SETS of k disjoint index pairs, and $E^{(k)\mu}{}_\mu$ (no sum) is $-\frac{1}{2}$ times the same sum over the pairs that avoid μ:

$$L_k = 2^k k! \sum_{\{p_1,\ldots,p_k\}} K_{p_1}\cdots K_{p_k}, \qquad E^{(k)\mu}{}_\mu = -2^{k-1}k! \sum_{\{p_1,\ldots,p_k\}\not\ni\mu} K_{p_1}\cdots K_{p_k}.$$

Counting the pairs among $x_1,\ldots,x_7$ (3 inside ordinary space, 6 inside the auxiliary space, 12 mixed) and the sets of two disjoint pairs (18 of type ordinary–auxiliary, 12 ordinary–mixed, 3 auxiliary–auxiliary, 36 auxiliary–mixed, 36 mixed–mixed) gives

$$G^0{}_0 = -\big(3K_{12} + 6K_{45} + 12K_{14}\big) = 3H_a^2 + 12H_aH_b + 6H_b^2,$$
$$E^{(2)0}{}_0 = -4\big(12H_a^3H_b + 54H_a^2H_b^2 + 36H_aH_b^3 + 3H_b^4\big) = -12H_b\big(4H_a^3 + 18H_a^2H_b + 12H_aH_b^2 + H_b^3\big).$$

Kasner exponents. With $a = x_0^p$, $b = x_0^q$: $H_a = p/x_0$, $H_a' = -p/x_0^2$ (likewise for b), and $x_0^{2k}E^{(k)}$ is a polynomial in p, q. Einstein: $3p + 4q = 1$, $3p^2 + 4q^2 = 1$. Gauss–Bonnet: $3p + 4q = 3$ and $-203p^3 + 81p^2 + 351p + 27 = 0$. Cubic: $3p + 4q = 5$ and $(p, q) = (-15/7, 20/7)$, besides the degenerate families $q = 0$ and $p = 0$.

Spin connection of the cosmological metric. From $\omega_\mu{}^A{}_B = e^A{}_\nu(\partial_\mu e_B{}^\nu + \{{}^\nu_{\mu\lambda}\}e_B{}^\lambda)$ with $\{{}^0_{ii}\} = aa'$, $\{{}^i_{0i}\} = a'/a$: $\omega_i{}^0{}_i = a'$, hence $\Omega_i = \frac{1}{4}(\omega_{i0i}\Gamma^0\Gamma^i + \omega_{ii0}\Gamma^i\Gamma^0) = \frac{a'}{2}\Gamma_i\Gamma_0$.

Static radial metric (Cartan). $e^0 = \sqrt{f}dx_0$, $e^1 = \sqrt{h}dr$, $e^k = r dx^k$; $de^0 = \frac{f'}{2\sqrt{f}}dr\wedge dx_0$, $de^k = dr\wedge dx^k$, $de^1 = 0$; Cartan's equation $de^A + \omega^A{}_B\wedge e^B = 0$ gives

$$\omega^0{}_1 = \frac{f'}{2\sqrt{fh}}\,dx_0, \quad \omega^k{}_1 = \frac{dx^k}{\sqrt{h}}, \quad \Omega_0 = \frac{f'}{4\sqrt{f}\sqrt{h}}\Gamma_1\Gamma_0, \quad \Omega_k = \frac{1}{2\sqrt{h}}\Gamma_1\Gamma_k, \quad \Gamma^\mu\Omega_\mu = \frac{1}{\sqrt{h}}\Big(\frac{f'}{4f} + \frac{3}{r}\Big)\Gamma^1.$$

Exponential vacuum solutions. For $a = e^{Hx_0}$, $b = e^{q_H Hx_0}$ the vacuum equations are linear in (w_2, w_3, Λ), and the divergence identity $\frac{d}{dx_0}E^{(k)0}{}_0 + 3H_a(E^{(k)0}{}_0 - E^{(k)1}{}_1) + 4H_b(E^{(k)0}{}_0 - E^{(k)4}{}_4) = 0$ with constant entries makes them of rank 2. For $q_H = -1$: $w_2 = \Lambda/4 - w_1/2$, $w_3 = w_1/24 - \Lambda/36$.

de Sitter pair creation. For $i\dot{c} = (ke^{-Hx_0}\sigma_1 + M\sigma_3)c$ the change of variable $z \propto e^{-Hx_0}$ gives Whittaker functions; their connection formula yields $|\beta|^2 = 1/(e^{2\pi M/H} + 1)$, checked numerically below.

5. Verification

Evaluation of 7+1.nb. The notebook was generated by `tools/make_7p1_nb.py` (which executes `tools/nb_part_gravity.py`, `tools/nb_part_ks_gravity.py`, `tools/nb_part_sm.py`) and evaluated by `tools/run_eval.wl` in a fresh kernel of Wolfram 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026); seconds 225.3, message cells 0, finished Tue 27 Oct 2026 14:15:35. The final section of the notebook lists 346 checks; all passed: True.

The checks of sections 11–14 of 7+1.nb:

#	check	result	section
1	cosmological metric: sectional curvatures K_0i = -(Ha'+Ha^2), K_0k = -(Hb'+Hb^2), K_ij = -Ha^2, K_kl = -Hb^2, K_ik = -Ha Hb	True	11.2
2	cosmological metric: R^cd_ab is non-zero only for {c,d} = {a,b}	True	11.2
3	Einstein tensor: G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2, G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb, G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb	True	11.2
4	scalar curvature R = -2 (3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 Ha Hb)	True	11.2
5	inhomogeneous metric, exactly at (x0, x4) = (1/2, 1/3): R_rsmn = -R_srmn = -R_rsnm = R_mnrs and R^r_[smn] = 0 (first Bianchi identity)	True	11.3
6	inhomogeneous metric: contracted Bianchi identity nabla_mu G^mu_nu == 0, exactly at (x0, x4) = (1/2, 1/3)	True	11.3
7	cosmological metric: contracted Bianchi identity nabla_mu G^mu_nu = 0 (arbitrary a, b)	True	11.3
8	generalized Kronecker delta: contraction identity delta^{m1..mp r}_{n1..np r} = (8-p) delta^{m1..mp}_{n1..np} (p = 1..4, 60 random index sets each)	True	12.1
9	generalized Kronecker delta with 9 upper indices vanishes in 8 dimensions (100 random index sets)	True	12.1
10	xAct: Lovelock1 == R_a^b - (1/2) delta_a^b R, the Einstein tensor (abstract indices)	True	12.3
11	xAct: trace of Lovelock2 == -2 L_2 with L_2 = R^2 - 4 R_cd R^cd + R_cdfi R^cdfi (abstract indices)	True	12.3
12	xAct, cosmological metric: Lovelock1 == EinsteinCD == the Einstein tensor of the curvature engine of section 11.1	True	12.4
13	pair-set algorithm == brute-force generalized-delta contraction for E^(1) (cosmological metric)	True	12.5
14	pair-set algorithm == xAct (section 12.4) for E^(1), E^(2), E^(3) of the cosmological metric (two independent computations)	True	12.5
15	E^(1) == Einstein tensor G and L_1 == R (cosmological metric)	True	12.6
16	trace identity E^(k)mu_mu == -((8-2k)/2) L_k for k = 0..4	True	12.6
17	E^(4) == 0 identically in 8 dimensions, while the Euler density L_4 != 0	True	12.6
18	every E^(k) is diagonal, E^(k)1_1 = E^(k)2_2 = E^(k)3_3 and E^(k)4_4 = ... = E^(k)7_7	True	12.6
19	E^(2) == Lanczos tensor and L_2 == R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric)	True	12.6
20	E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2 Hb + 12 Ha Hb^2 + Hb^3) and E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb)	True	12.6
21	divergence-free: d0 E^0_0 + 3Ha (E^0_0 - E^1_1) + 4Hb (E^0_0 - E^4_4) == 0 for k = 1, 2, 3 (arbitrary a, b)	True	12.6
22	variation of nL a^3 b^4 L_k: EL_nL = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4 E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4 (k = 1..4)	True	12.6
23	inhomogeneous metric: nabla_mu E^(k)mu_nu == 0 EXACTLY at (x0, x4) = (1/2, 1/3) for k = 1, 2, 3	True	12.7
24	inhomogeneous metric: E^(1) == G at the point, and E^(k)_mu nu symmetric (k = 1, 2, 3)	True	12.7
25	inhomogeneous metric: E^(2) == Lanczos tensor at the point (exact)	True	12.7
26	pp-wave: E^(1)_mu nu == +(1/2)(d2^2 + d3^2) fPP l_mu l_nu	True	12.8
27	pp-wave: E^(2) == E^(3) == 0 for every profile fPP	True	12.8
28	pp-wave: xAct (section 12.4) == pair-set algorithm for E^(1), E^(2), E^(3) (a metric that is not diagonal; arbitrary profile fPP)	True	12.8
29	exponential expansion a = e^{H x0}, b = e^{qH H x0}: xAct Lovelock_k == the closed forms of section 12.6 with Ha = H, Hb = qH H, Ha' = Hb' = 0 (k = 1, 2, 3)	True	12.9
30	exponential expansion: three distinct vacuum equations, related by 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) == 0	True	12.9
31	exponential vacuum solutions: (w2, w3) solve all 64 components for every qH (generic qH) and every Lambda	True	12.9
32	eternal inflation (a = e^{H x0}) with deflation (b = e^{-H x0}): w2 == Lambda/4 - w1/2, w3 == w1/24 - Lambda/36; for Lambda = 0: w2/w1 == -1/2, w3/w1 == 1/24	True	12.9
33	cosmological metric: Omega_0 == 0, Omega_i == (a'/2) Gamma17[i].Gamma17[0], Omega_k == (b'/2) Gamma17[k].Gamma17[0]	True	13.2

#	check	result	section
34	cosmological metric: Gamma^mu Omega_mu == (3a'/(2a) + 2b'/b) Gamma17[0]	True	13.2
35	current identity: d_mu(sqrt|g| psibar Gamma^mu psi) == sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi] (psi, psibar arbitrary)	True	13.2
36	Lichnerowicz formula: (Gamma^mu D_mu)^2 psi == (g^mu nu nabla_mu D_nu - R/4) psi	True	13.2
37	rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi == (a^3 b^4)^(-1/2) Gamma^mu d_mu chi	True	13.2
38	mode equation: Gamma17[0].(i Gamma^mu D_mu - m) psi == (a^3 b^4)^(-1/2) e^{i(p.x+q.y)} (i dc/dx0 - h(x0) c) for psi = (a^3 b^4)^(-1/2) e^{i(p.x+q.y)} c(x0)	True	13.2
39	conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) == C^(-9/2) Gamma17up[A] d_A f for g = C(x0)^2 eta17	True	13.4
40	Cartan's first structure equation de^A + omega^A_B ^ e^B == 0 (no torsion): cosmological, pp-wave and inhomogeneous frames	True	13.5
41	structure coefficients: omega_C,AB == -(1/2)(C_ACB + C_BAC - C_CBA) with C^A_BC = e_B^mu e_C^nu (d_mu e^A_nu - d_nu e^A_mu): cosmological and inhomogeneous frames	True	13.5
42	static radial metric: Cartan's first structure equation and omega_C,AB == -(1/2)(C_ACB + C_BAC - C_CBA)	True	13.5
43	static radial metric: Omega_0 == (f'/(4 Sqrt[f] Sqrt[h])) Gamma17[1].Gamma17[0], Omega_1 == 0, Omega_k == (1/(2 Sqrt[h])) Gamma17[1].Gamma17[k] (k = 2..7)	True	13.5
44	static radial metric: Gamma^mu Omega_mu == (1/Sqrt[h]) (f'/(4 f) + 3/r) Gamma17up[1]	True	13.5
45	static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection, Gamma^mu D_mu psi == f^(-1/4) r^(-3) Gamma^mu d_mu chi	True	13.5
46	U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) == e^{i q alpha} Gamma^mu (D_mu - i q A_mu) psi with A'_mu = A_mu + d_mu alpha	True	13.5
47	current identity with the gauge field: d_mu(sqrt|g| psibar Gamma^mu psi) == sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu	True	13.5
48	Kasner metric: the Einstein tensor computed directly from the metric equals the closed form (x0^2 G^mu_mu, mu = 0, 1, 4)	True	14.1
49	Einstein Kasner: (p, q) = ((1+2Sqrt[2])/7, (2-3Sqrt[2])/14) and ((1-2Sqrt[2])/7, (2+3Sqrt[2])/14) solve E^(1) == 0	True	14.1
50	Gauss-Bonnet Kasner: 3 real roots of the cubic; (p, (3-3p)/4) solve E^(2) == 0	True	14.1
51	cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) == 0; E^(3) == 0 identically for q = 0 and for p = 0	True	14.1
52	2x2 reduction: A, B, betaD square to ID16 and anticommute; Zc = A.B.betaD: Zc.Zc == -ID16, Transpose[Zc] == -Zc, Zc commutes with A, B, betaD	True	14.2
53	Kasner pair creation: |alpha|^2 + |beta|^2 == 1 (to 10^-8) for (|p|, |q|) = (1,1), (2,1), (1,3)	True	14.3
54	Kasner scaling: (|p|, |q|) = (1.4302.., 2.5) has K = 1 and the same |beta|^2 as (1, 1) (to 10^-6)	True	14.3
55	16x16 evolution == 4 copies of the 2x2 evolution with +m sigma3 and 4 with -m sigma3 (Kasner, m = 0.3, |p| = 1, |q| = 1.5, x0 from 0.05 to 40): average |beta|^2 agrees to 10^-8; unitarity; the two classes differ	True	14.3
56	isotropic expansion, massless: no pair creation (|beta|^2 < 10^-20)	True	14.3
57	static auxiliary space (Hb = 0): E^(1) = (3Ha^2, 2Ha' + 3Ha^2, 3Ha' + 6Ha^2), E^(2) = (0, 0, -12 Ha^2 (Ha' + Ha^2)), E^(3) = 0	True	14.4
58	8D de Sitter: E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (all diagonal entries)	True	14.5
59	de Sitter over a static auxiliary space: E^(1) = (3, 3, 6) H^2, E^(2) = (0, 0, -12) H^4, E^(3) = 0	True	14.5
60	de Sitter pair creation: |beta|^2 == 1/(Exp[2 Pi M/H] + 1) to 10^-7, M/H = 0.25, 0.5, 1	True	14.5
61	integrator benchmark (Landau-Zener): |beta|^2 == Exp[-Pi kappa^2] to 10^-6	True	14.5
62	cosmological metric: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True	13.1
63	cosmological metric: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True	13.1
64	cosmological metric: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True	13.1
65	cosmological metric: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True	13.1
66	cosmological metric: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True	13.1
67	cosmological metric: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True	13.1
68	cosmological metric: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True	13.1
69	cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True	13.1
70	cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True	13.1

#	check	result	section
71	pp-wave: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True	13.1
72	pp-wave: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True	13.1
73	pp-wave: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True	13.1
74	pp-wave: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True	13.1
75	pp-wave: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True	13.1
76	pp-wave: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True	13.1
77	pp-wave: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True	13.1
78	pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True	13.1
79	pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True	13.1
80	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True	13.1
81	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True	13.1
82	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True	13.1
83	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True	13.1
84	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True	13.1
85	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True	13.1
86	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True	13.1
87	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True	13.1
88	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True	13.1
89	static radial metric: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True	13.1
90	static radial metric: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True	13.1
91	static radial metric: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True	13.1
92	static radial metric: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True	13.1
93	static radial metric: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True	13.1
94	static radial metric: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True	13.1
95	static radial metric: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True	13.1
96	static radial metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True	13.1
97	static radial metric: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True	13.1

The independent Python program tools/ref_gravity.py (its own curvature and Lovelock engine tools/lovelock71.py; the Dirac matrices rebuilt from the split octonions of 4+4.nb by tools/ref71.py). Its boolean checks:

#	check	result
1	cosmological metric: sectional curvatures R^{AB}_{AB} = -(Ha'+Ha^2), -(Hb'+Hb^2), -Ha^2, -Hb^2, -Ha Hb	True
2	cosmological metric: Riemann R^{cd}_{ab} non-zero only for {c,d} = {a,b}	True
3	G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2	True
4	G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb	True
5	G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb	True
6	scalar curvature R = -2(3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 HaHb)	True
7	E^(1) = Einstein tensor G (cosmological metric)	True
8	L_1 = R	True
9	E^(0) = -delta/2	True

#	check	result
10	trace identity E^(k)mu_mu = -(8-2k)/2 L_k, k = 0..4	True
11	E^(4) = 0 identically in 8 dimensions, but L_4 (Euler density) is not 0	True
12	E^(k) diagonal: E^(k)mu_nu = 0 for mu != nu, and E^(k)1_1 = E^(k)2_2 = E^(k)3_3, E^(k)4_4 = ... = E^(k)7_7	True
13	E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2Hb + 12HaHb^2 + Hb^3)	True
14	E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb)	True
15	E^(2) = Lanczos tensor 2(R R_mn - 2R_ma R^a_n - 2R^ab R_manb + R_m^abc R_nabc) - g_mn L_2/2, and L_2 = R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric)	True
16	divergence-free E^(k), k = 1, 2, 3 (cosmological metric, arbitrary a, b)	True
17	variation of sqrt|g| L_k (lapse N, a, b): EL_N = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4 E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4, k = 1..4	True
18	inhomogeneous metric (N, A, B of x0 and x4): E^(1) = G at the point	True
19	inhomogeneous metric: nabla_mu E^(k)mu_nu = 0 exactly at x0 = 1/2, x4 = 1/3, k = 1, 2, 3	True
20	inhomogeneous metric: E^(k)_mu nu symmetric, k = 1, 2, 3	True
21	pp-wave: E^(1)_mu nu = +(1/2) (d2^2 + d3^2) F l_mu l_nu, l = dx0 - dx1	True
22	pp-wave: E^(2) = E^(3) = 0 for every profile F	True
23	Einstein Kasner: p = (1+-2 sqrt2)/7, q = (2-+3 sqrt2)/14 solve E^(1) = 0; 3p+4q = 1, 3p^2+4q^2 = 1	True
24	Gauss-Bonnet Kasner: the three (p, (3-3p)/4) solve E^(2) = 0, sum of the 7 exponents = 3	True
25	cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) = 0, 3p + 4q = 5; degenerate families q = 0 and p = 0	True
26	Einstein Kasner is not a cubic-Lovelock vacuum (E^(3) != 0)	True
27	8D de Sitter (a = b = e^{H x0}): E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (times delta)	True
28	4D de Sitter on a static T^4 (a = e^{H x0}, b = b0): E^(1) = diag(3,3,6) H^2, E^(2) = diag(0,0,-12) H^4, E^(3) = 0	True
29	static T^4 (Hb = 0): E^(1) = (3Ha^2, 2Ha'+3Ha^2, 3Ha'+6Ha^2), E^(2) = (0, 0, -12 Ha^2 (Ha'+Ha^2)), E^(3) = 0	True
30	exponential expansion a = e^{H x0}, b = e^{q H x0}: E^(k) (00, 11, 44) = closed forms with Ha = H, Hb = q H, Ha' = Hb' = 0 (k = 1, 2, 3)	True
31	exponential expansion: 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) = 0 identically (the three vacuum equations have rank 2)	True
32	exponential vacuum solutions: one solution (w2, w3) for given w1, Lambda, q, satisfying all three equations	True
33	eternal inflation a = e^{H x0} with deflation b = e^{-H x0}: w2 = Lambda/4 - w1/2, w3 = w1/24 - Lambda/36 (Lambda = 0: w2/w1 = -1/2, w3/w1 = 1/24)	True
34	cosmological metric: eta e e = g	True
35	cosmological metric: {Gamma_mu, Gamma_nu} = 2 g_mu nu	True
36	cosmological metric: Gamma_mu real	True
37	cosmological metric: omega_mu AB = -omega_mu BA	True
38	cosmological metric: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0	True
39	cosmological metric: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu	True
40	cosmological metric: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0	True
41	cosmological metric: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0	True
42	cosmological metric: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B	True
43	cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0	True
44	cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16	True
45	cosmological metric: Gamma^mu Omega_mu = (3a'/(2a) + 2b'/b) Gamma17[0]	True
46	cosmological metric: Omega_i = (a'/2) Gamma17[i] Gamma17[0] (i = 1..3), Omega_k = (b'/2) Gamma17[k] Gamma17[0] (k = 4..7), Omega_0 = 0	True
47	pp-wave: eta e e = g	True
48	pp-wave: {Gamma_mu, Gamma_nu} = 2 g_mu nu	True
49	pp-wave: Gamma_mu real	True
50	pp-wave: omega_mu AB = -omega_mu BA	True
51	pp-wave: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0	True
52	pp-wave: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu	True
53	pp-wave: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0	True
54	pp-wave: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0	True
55	pp-wave: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B	True
56	pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0	True
57	pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16	True
58	inhomogeneous metric N, A, B (x0, x4): eta e e = g	True
59	inhomogeneous metric N, A, B (x0, x4): {Gamma_mu, Gamma_nu} = 2 g_mu nu	True

#	check	result
60	inhomogeneous metric N, A, B (x0, x4): Gamma_mu real	True
61	inhomogeneous metric N, A, B (x0, x4): omega_mu AB = -omega_mu BA	True
62	inhomogeneous metric N, A, B (x0, x4): vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0	True
63	inhomogeneous metric N, A, B (x0, x4): Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu	True
64	inhomogeneous metric N, A, B (x0, x4): total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0	True
65	inhomogeneous metric N, A, B (x0, x4): constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0	True
66	inhomogeneous metric N, A, B (x0, x4): spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B	True
67	inhomogeneous metric N, A, B (x0, x4): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0	True
68	inhomogeneous metric N, A, B (x0, x4): (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16	True
69	Cartan's first structure equation de^A + omega^A_B ^ e^B = 0 (torsion-free) for the cosmological frame	True
70	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): eta e e = g	True
71	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): {Gamma_mu, Gamma_nu} = 2 g_mu nu	True
72	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Gamma_mu real	True
73	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): omega_mu AB = -omega_mu BA	True
74	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0	True
75	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu	True
76	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0	True
77	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0	True
78	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B	True
79	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0	True
80	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16	True
81	Cartan's first structure equation for the static radial metric	True
82	structure coefficients: de^A = (1/2) C^A_BC e^B ^ e^C and omega_C,AB (form index C) = -(1/2)(C_ACB + C_BAC - C_CBA) (static radial metric)	True
83	static radial metric: Omega_0 = (f'/(4 sqrt(f h))) Gamma17[1] Gamma17[0], Omega_1 = 0, Omega_k = (1/(2 sqrt h)) Gamma17[1] Gamma17[k] (k = 2..7)	True
84	static radial metric: Gamma^mu Omega_mu = (1/sqrt h)(f'/(4f) + 3/r) Gamma17up[1], so (i Gamma^mu D_mu - m) psi = [i Gamma17up[0] d0/sqrt f + i Gamma17up[1] (d_r + f'/(4f) + 3/r)/sqrt h + sum_k i Gamma17up[k] d_k/r - m] psi	True
85	static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection: Gamma^mu D_mu psi = f^(-1/4) r^(-3) Gamma^mu d_mu chi	True
86	U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) = e^{i q alpha} Gamma^mu (D_mu - i q A_mu) psi with A' = A + d alpha	True
87	current identity with the gauge field: d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu	True
88	current identity d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi]	True
89	Lichnerowicz: (Gamma^mu D_mu)^2 psi = (g^mu nu nabla_mu D_nu - R/4) psi	True
90	rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi = (a^3 b^4)^(-1/2) Gamma^mu d_mu chi	True
91	mode equation: (i Gamma^mu D_mu - m) psi = 0 <=> i dc/dx0 = h(x0) c, h = alpha.(p/a, q/b) + beta m	True
92	conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) = C^(-9/2) Gamma^A d_A f for g = C(x0)^2 eta	True
93	2x2 reduction: A = alpha.n_a, B = alpha.n_b, beta: squares 1, mutually anticommuting; Z = A B beta: Z^2 = -1, Z^T = -Z, commutes with A, B, beta	True
94	Kasner: |alpha|^2 + |beta|^2 = 1 (to 1e-8)	True
95	Kasner scaling: (|p|, |q|) = (1.430..., 2.5) has K = 1 and the same |beta|^2 as (1, 1) (to 1e-6)	True
96	16x16 evolution = 4 copies of the 2x2 evolution with +m sigma3 and 4 copies with -m sigma3: average agrees (to 1e-8), unitarity; the two sectors differ	True
97	isotropic expansion, massless: no pair creation (|beta|^2 < 1e-20)	True

#	check	result
98	de Sitter: |beta|^2 = 1/(exp(2 pi m/H) + 1) (Fermi-Dirac at the Gibbons-Hawking temperature H/2pi), to 1e-7	True
99	integrator benchmark: Landau-Zener |beta|^2 = exp(-pi kappa^2) (to 1e-6)	True

Numbers computed by both programs:

quantity	7+1.nb	Python
Kasner $\|\beta\|^2$ at $(\|\vec{p}\|, \|\vec{q}\|) = (1, 1)$	0.010749346	0.010749346
Kasner $\|\beta\|^2$ at $(\|\vec{p}\|, \|\vec{q}\|) = (2, 1)$	0.00013294108	0.00013294108
Kasner $\|\beta\|^2$ at $(\|\vec{p}\|, \|\vec{q}\|) = (1, 3)$	0.067442022	0.067442022
de Sitter $\|\beta\|^2$, $M/H = 0.25$	0.172102919	—
de Sitter $\|\beta\|^2$, $M/H = 0.5$	0.0414238845	—
de Sitter $\|\beta\|^2$, $M/H = 1$	0.00186397746	—

SHA-256 of the files (at the time this document was generated):

```
7+1.nb                          060942488caf4f45120ffd37be0123e96992a31cd7fdbde09ee4dc4ced4f5
    ...d3a
tools/make_7p1_nb.py            2fac94dfd0c9fcf3aed8d6e90a77fbbb8ab1c8450819ff591b7e5cf7b5787
    ...d6f
tools/nb_part_gravity.py        55d16f735e16df7a7cd2aa70d322969bc6cf82c777965f112b5a9acf46a6a
    ...d5f
tools/nb_part_ks_gravity.py     886814fedc4cd015c63b950ee83724025ba21fd6c3e56b68dcb7f2f07afdc
    ...205
tools/nb_part_sm.py             20782f1fe557fd3f3dea7417e89ecd42a1602f7a48599fcdcfd0cff77f120
    ...ecc
tools/lovelock71.py             d93ecbd0f82adb972218d59dfe32999a4af944945ca9758607f3a633e1af9
    ...5a1
tools/ref_gravity.py            2bfdc087ed4dc7d84258e04449d11141b415f54b992e65d42fa42cace08ff
    ...6bd
tools/ref_ks_gravity.py         8d082985c5161254d7c6294c0310f7807aa037bff27a8cd3ec4d3f8a9f6e0
    ...550
tools/ref_sm.py                 d36d23f472c8339d80f1d0b1eb46c509398562053ead7033d96b054782e73
    ...b49
tools/ref71.py                  4ac783b1ab221317387f2a74e0be60f114fd64940fe3ea96af79b33ee7dcb
    ...8b0
tools/ref44.py                  ab100c3ebcc055fbe59c132a02f80ecb17630e934e99d59f88e919ec8064f
    ...af5
```

Output of tools/ref_gravity.py (independent checks of sections 11-14; exit code 0) (complete):

```
cosmological metric: sectional curvatures R^{AB}_{AB} = -(Ha'+Ha^2), -(Hb'+Hb^2), -Ha^2, -Hb^2,
    ...-Ha Hb : True
cosmological metric: Riemann R^{cd}_{ab} non-zero only for {c,d} = {a,b} : True
G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2 : True
G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb : True
G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb : True
scalar curvature R = -2(3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 HaHb) : True
E^(1) = Einstein tensor G (cosmological metric) : True
L_1 = R : True
E^(0) = -delta/2 : True
trace identity E^(k)mu_mu = -(8-2k)/2 L_k, k = 0..4 : True
E^(4) = 0 identically in 8 dimensions, but L_4 (Euler density) is not 0 : True
E^(k) diagonal: E^(k)mu_nu = 0 for mu != nu, and E^(k)1_1 = E^(k)2_2 = E^(k)3_3, E^(k)4_4 = ...
    ...= E^(k)7_7 : True
E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2Hb + 12HaHb^2 + Hb^3) : True
E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb) : True
```

```
E^(2) = Lanczos tensor 2(R R_mn - 2R_ma R^a_n - 2R^ab R_manb + R_m^abc R_nabc) - g_mn L_2/2,
    ...and L_2 = R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric) : True
closed forms : {(1, 0): 3*H_a**2 + 12*H_a*H_b + 6*H_b**2, (1, 1): 3*H_a**2 + 8*H_a*H_b +
    ...10*H_b**2 + 2*dH_a + 4*dH_b, (1, 4): 6*H_a**2 + 9*H_a*H_b + 6*H_b**2 + 3*dH_a + 3*dH_b,
    ...(2, 0): -48*H_a**3*H_b - 216*H_a**2*H_b**2 - 144*H_a*H_b**3 - 12*H_b**4, (2, 1):
    ...-32*H_a**3*H_b - 136*H_a**2*H_b**2 - 16*H_a**2*dH_b - 192*H_a*H_b**3 - 32*H_a*H_b*dH_a -
    ...96*H_a*H_b*dH_b - 60*H_b**4 - 48*H_b**2*dH_a - 48*H_b**2*dH_b, (2, 4): -12*H_a**4 -
    ...108*H_a**3*H_b - 180*H_a**2*H_b**2 - 12*H_a**2*dH_a - 36*H_a**2*dH_b - 108*H_a*H_b**3 -
    ...72*H_a*H_b*dH_a - 72*H_a*H_b*dH_b - 12*H_b**4 - 36*H_b**2*dH_a - 12*H_b**2*dH_b, (3, 0):
    ...1440*H_a**3*H_b**3 + 1080*H_a**2*H_b**4, (3, 1): 576*H_a**3*H_b**3 + 1368*H_a**2*H_b**4
    ...+ 864*H_a**2*H_b**2*dH_b + 576*H_a*H_b**5 + 576*H_a*H_b**3*dH_a + 576*H_a*H_b**3*dH_b +
    ...144*H_b**4*dH_a, (3, 4): 648*H_a**4*H_b**2 + 1224*H_a**3*H_b**3 + 432*H_a**3*H_b*dH_b +
    ...648*H_a**2*H_b**4 + 648*H_a**2*H_b**2*dH_a + 648*H_a**2*H_b**2*dH_b +
    ...432*H_a*H_b**3*dH_a}
divergence-free E^(k), k = 1, 2, 3 (cosmological metric, arbitrary a, b) : True
variation of sqrt|g| L_k (lapse N, a, b): EL_N = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4
    ...E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4, k = 1..4 : True
inhomogeneous metric: E^(3)0_4 at the point (exact) :
    ...353743050254866364332989413672793316453216419840000/24577736888913221711303629009057562607
    ...63178144678501
inhomogeneous metric (N, A, B of x0 and x4): E^(1) = G at the point : True
inhomogeneous metric: nabla_mu E^(k)mu_nu = 0 exactly at x0 = 1/2, x4 = 1/3, k = 1, 2, 3 : True
inhomogeneous metric: E^(k)_mu nu symmetric, k = 1, 2, 3 : True
pp-wave: E^(1)_mu nu = +(1/2) (d2^2 + d3^2) F l_mu l_nu, l = dx0 - dx1 : True
pp-wave: E^(2) = E^(3) = 0 for every profile F : True
Kasner E^(k) components x0^(2k) (00, 11, 44) : {1: [3*(p**2 + 4*p*q + 2*q**2), 3*p**2 + 8*p*q -
    ...2*p + 10*q**2 - 4*q, 3*(2*p**2 + 3*p*q - p + 2*q**2 - q)], 2: [-12*q*(4*p**3 + 18*p**2*q
    ...+ 12*p*q**2 + q**3), -4*q*(8*p**3 + 34*p**2*q - 12*p**2 + 48*p*q**2 - 36*p*q + 15*q**3 -
    ...12*q**2), -12*(p**4 + 9*p**3*q - p**3 + 15*p**2*q**2 - 9*p**2*q + 9*p*q**3 - 9*p*q**2 +
    ...q**4 - q**3)], 3: [360*p**2*q**3*(4*p + 3*q), 72*p*q**3*(8*p**2 + 19*p*q - 20*p + 8*q**2
    ...- 10*q), 72*p**2*q**2*(9*p**2 + 17*p*q - 15*p + 9*q**2 - 15*q)]}
Einstein Kasner: p = (1+-2 sqrt2)/7, q = (2-+3 sqrt2)/14 solve E^(1) = 0; 3p+4q = 1, 3p^2+4q^2
    ...= 1 : True
Gauss-Bonnet Kasner cubic (64 x e4 at q = (3-3p)/4) : -203*p**3 + 81*p**2 + 351*p + 27
Gauss-Bonnet Kasner exponents p (three real roots) : (-1.0835017876121057424239762947 3,
    ...-0.07863105841892776622786105210 99, 1.561147624356156661361196952 75)
Gauss-Bonnet Kasner: the three (p, (3-3p)/4) solve E^(2) = 0, sum of the 7 exponents = 3 : True
cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) = 0, 3p + 4q = 5; degenerate
    ...families q = 0 and p = 0 : True
Einstein Kasner is not a cubic-Lovelock vacuum (E^(3) != 0) : True
8D de Sitter (a = b = e^{H x0}): E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (times
    ...delta) : True
4D de Sitter on a static T^4 (a = e^{H x0}, b = b0): E^(1) = diag(3,3,6) H^2, E^(2) =
    ...diag(0,0,-12) H^4, E^(3) = 0 : True
static T^4 (Hb = 0): E^(1) = (3Ha^2, 2Ha'+3Ha^2, 3Ha'+6Ha^2), E^(2) = (0, 0, -12 Ha^2
    ...(Ha'+Ha^2)), E^(3) = 0 : True
exponential expansion a = e^{H x0}, b = e^{q H x0}: E^(k) (00, 11, 44) = closed forms with Ha =
    ...H, Hb = q H, Ha' = Hb' = 0 (k = 1, 2, 3) : True
exponential expansion: 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) = 0 identically (the three
    ...vacuum equations have rank 2) : True
exponential vacuum solutions: one solution (w2, w3) for given w1, Lambda, q, satisfying all
    ...three equations : True
eternal inflation a = e^{H x0} with deflation b = e^{-H x0}: w2 = Lambda/4 - w1/2, w3 = w1/24 -
    ...Lambda/36 (Lambda = 0: w2/w1 = -1/2, w3/w1 = 1/24) : True
NUM exponential vacuum solution (w2, w3) for general q : ((-2*Lambda*q - 3*Lambda + 12*q**3*w1
    ...+ 27*q**2*w1 + 22*q*w1 + 9*w1)/(4*q*(6*q**4 + 36*q**3 + 66*q**2 + 51*q + 16)),
    ...(-3*Lambda*q**2 - 6*Lambda*q - Lambda + 15*q**4*w1 + 36*q**3*w1 + 33*q**2*w1 + 18*q*w1 +
    ...3*w1)/(72*q**3*(6*q**4 + 36*q**3 + 66*q**2 + 51*q + 16)))
NUM eq00, eq11, eq44 for q = -1 : (-Lambda - 3*w1 - 36*w2 - 360*w3, -Lambda + 5*w1 + 28*w2 +
    ...216*w3, -Lambda + 3*w1 + 12*w2 + 72*w3)
cosmological metric: eta e e = g : True
cosmological metric: {Gamma_mu, Gamma_nu} = 2 g_mu nu : True
cosmological metric: Gamma_mu real : True
cosmological metric: omega_mu AB = -omega_mu BA : True
cosmological metric: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu =
```

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    ...0 : True
cosmological metric: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu : True
cosmological metric: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu
    ...nu Gamma_l + [Omega_mu, Gamma_nu] = 0 : True
cosmological metric: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 :
    ...True
cosmological metric: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B : True
cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0 : True
cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 : True
cosmological metric: Gamma^mu Omega_mu = (3a'/(2a) + 2b'/b) Gamma17[0] : True
cosmological metric: Omega_i = (a'/2) Gamma17[i] Gamma17[0] (i = 1..3), Omega_k = (b'/2)
    ...Gamma17[k] Gamma17[0] (k = 4..7), Omega_0 = 0 : True
pp-wave: eta e e = g : True
pp-wave: {Gamma_mu, Gamma_nu} = 2 g_mu nu : True
pp-wave: Gamma_mu real : True
pp-wave: omega_mu AB = -omega_mu BA : True
pp-wave: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0 : True
pp-wave: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu : True
pp-wave: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l
    ...+ [Omega_mu, Gamma_nu] = 0 : True
pp-wave: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 : True
pp-wave: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B : True
pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0 : True
pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 : True
inhomogeneous metric N, A, B (x0, x4): eta e e = g : True
inhomogeneous metric N, A, B (x0, x4): {Gamma_mu, Gamma_nu} = 2 g_mu nu : True
inhomogeneous metric N, A, B (x0, x4): Gamma_mu real : True
inhomogeneous metric N, A, B (x0, x4): omega_mu AB = -omega_mu BA : True
inhomogeneous metric N, A, B (x0, x4): vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l +
    ...omega_mu^A_B e^B_nu = 0 : True
inhomogeneous metric N, A, B (x0, x4): Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] =
    ...-Omega_mu : True
inhomogeneous metric N, A, B (x0, x4): total covariant derivative of the frame matrices: d_mu
    ...Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0 : True
inhomogeneous metric N, A, B (x0, x4): constant frame matrices: omega_mu^A_B Gamma^B +
    ...[Omega_mu, Gamma^A] = 0 : True
inhomogeneous metric N, A, B (x0, x4): spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A
    ...Gamma^B : True
inhomogeneous metric N, A, B (x0, x4): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0
    ...: True
inhomogeneous metric N, A, B (x0, x4): (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 : True
Cartan's first structure equation de^A + omega^A_B ^ e^B = 0 (torsion-free) for the
    ...cosmological frame : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): eta e e = g : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): {Gamma_mu, Gamma_nu} =
    ...2 g_mu nu : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Gamma_mu real : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): omega_mu AB =
    ...-omega_mu BA : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): vielbein postulate
    ...d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Omega_mu real,
    ...Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): total covariant
    ...derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu,
    ...Gamma_nu] = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): constant frame
    ...matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): spin curvature F_mu nu
    ...= (1/4) R_AB mu nu Gamma^A Gamma^B : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Bianchi identity d_l
    ...F_mn + [Omega_l, F_mn] + cyclic = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): (1/2) Gamma^mu
    ...Gamma^nu F_mu nu = -(R/4) 1_16 : True
Cartan's first structure equation for the static radial metric : True
```

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structure coefficients: de^A = (1/2) C^A_BC e^B ^ e^C and omega_C,AB (form index C) =
    ...-(1/2)(C_ACB + C_BAC - C_CBA) (static radial metric) : True
static radial metric: Omega_0 = (f'/(4 sqrt(f h))) Gamma17[1] Gamma17[0], Omega_1 = 0, Omega_k
    ...= (1/(2 sqrt h)) Gamma17[1] Gamma17[k] (k = 2..7) : True
static radial metric: Gamma^mu Omega_mu = (1/sqrt h)(f'/(4f) + 3/r) Gamma17up[1], so (i
    ...Gamma^mu D_mu - m) psi = [i Gamma17up[0] d0/sqrt f + i Gamma17up[1] (d_r + f'/(4f) +
    ...3/r)/sqrt h + sum_k i Gamma17up[k] d_k/r - m] psi : True
static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection: Gamma^mu D_mu psi
    ...= f^(-1/4) r^(-3) Gamma^mu d_mu chi : True
U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) = e^{i q alpha} Gamma^mu
    ...(D_mu - i q A_mu) psi with A' = A + d alpha : True
current identity with the gauge field: d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu
    ...psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu : True
current identity d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu psibar) Gamma^mu psi +
    ...psibar Gamma^mu D_mu psi] : True
Lichnerowicz: (Gamma^mu D_mu)^2 psi = (g^mu nu nabla_mu D_nu - R/4) psi : True
rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi = (a^3
    ...b^4)^(-1/2) Gamma^mu d_mu chi : True
mode equation: (i Gamma^mu D_mu - m) psi = 0  <=>  i dc/dx0 = h(x0) c, h = alpha.(p/a, q/b) +
    ...beta m : True
conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) = C^(-9/2) Gamma^A d_A f for g = C(x0)^2 eta :
    ...True
2x2 reduction: A = alpha.n_a, B = alpha.n_b, beta: squares 1, mutually anticommuting; Z = A B
    ...beta: Z^2 = -1, Z^T = -Z, commutes with A, B, beta : True
Kasner (Einstein branch), massless: |beta|^2 for (|p|, |q|) = (1,1), (2,1), (1,3) :
    ...(np.float64(0.01074934616777871), np.float64(0.00013294108220674155),
    ...np.float64(0.06744202228393215))
Kasner scaling variable K = |p| lambda^(1-p), lambda = (|q|/|p|)^(1/(q-p)) : (1.0,
    ...3.11828865465698, 0.4946322370571032)
Kasner: |alpha|^2 + |beta|^2 = 1 (to 1e-8) : True
Kasner scaling: (|p|, |q|) = (1.430..., 2.5) has K = 1 and the same |beta|^2 as (1, 1) (to
    ...1e-6) : True
Kasner, m = 0.3, |p| = 1, |q| = 1.5, x0: 0.05 -> 40: |beta|^2 of the 16x16 evolution (average),
    ...of the sector Z = +i (+m sigma3) and of the sector Z = -i (-m sigma3) :
    ...(np.float64(0.027052647156543617), np.float64(0.02528659275098409),
    ...np.float64(0.028818701562187667))
16x16 evolution = 4 copies of the 2x2 evolution with +m sigma3 and 4 copies with -m sigma3:
    ...average agrees (to 1e-8), unitarity; the two sectors differ : True
isotropic expansion, massless: no pair creation (|beta|^2 < 1e-20) : True
far future (de Sitter, H = 1): |beta|^2 for m/H = 0.25, 0.5, 1 :
    ...(np.float64(0.1721029191835808), np.float64(0.04142388447252214),
    ...np.float64(0.001863977463333981))
de Sitter: |beta|^2 = 1/(exp(2 pi m/H) + 1) (Fermi-Dirac at the Gibbons-Hawking temperature
    ...H/2pi), to 1e-7 : True
integrator benchmark: Landau-Zener |beta|^2 = exp(-pi kappa^2) (to 1e-6) : True
boolean checks: 99   failed: 0
VERDICT: ALL CHECKS PASSED
```